

We will do the one-consumer case. Consider the sequence $\hat{\mathbf{x}}_i \rightarrow \mathbf{x}_i$ where $\mathbf{p} \cdot \hat{\mathbf{x}}_i \geq c$ for all parts in the sequence. We want to show that this inequality is preserved in the limit, i.e. that $\mathbf{p} \cdot \mathbf{x}_i \geq c$

Suppose toward a contradiction that instead $\mathbf{p} \cdot \mathbf{x}_i = d < c$.

From the definition of the limit of the sequence:

$$\lim_{\hat{\mathbf{x}}_i \rightarrow \mathbf{x}_i} \mathbf{p} \cdot \hat{\mathbf{x}}_i = d$$

implies that for $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. for every $\hat{\mathbf{x}}_i$, $0 < |\hat{\mathbf{x}}_i - \mathbf{x}_i| < \delta$ implies $|\mathbf{p} \cdot \hat{\mathbf{x}}_i - d| < \varepsilon$.

Because this holds for all $\varepsilon > 0$, let's choose $\varepsilon = c - d$. This means $\exists \delta > 0$ such that for all $\hat{\mathbf{x}}_i$ in the sequence, $0 < |\hat{\mathbf{x}}_i - \mathbf{x}_i| < \delta$ implies that $|\mathbf{p} \cdot \hat{\mathbf{x}}_i - d| < \varepsilon = c - d$.

But then:

$$-\varepsilon < \mathbf{p} \cdot \hat{\mathbf{x}}_i - d < c - d$$

Implies that $\mathbf{p} \cdot \hat{\mathbf{x}}_i < c$ for some parts of the sequence, a contradiction.